1. Using Structured Query Language (SQL)

1a. Explain the relationship between a database and SQL

2. Using Data Manipulation Language (DML) and Transaction Control Language (TCL)

2a. Describe the purpose of DML
2b. Use DML to manage data in tables
2c. Use TCL to manage transactions

3. Using Basic SELECT statements

3a. Build a SELECT statement to retrieve data from an Oracle Database table
3b. Use the WHERE clause to the SELECT statement to filter query results

4. Defining Table Joins

4a. Describe the different types of joins and their features
4b. Use joins to retrieve data from multiple tables
4c. Use self joins

5. Using Conversion Functions and Conditional Expressions

5a. Use the TO_CHAR, TO_NUMBER, and TO_DATE conversion functions
5b. Apply general functions and conditional expressions in a SELECT statement

6. Displaying Data from Multiple Tables

6a. Use SELECT statements to access data from more than one table using equijoins and nonequijoins
6b. Join a table to itself by using a self-join
6c. View data that generally does not meet a join condition by using outer joins

7. Using the Set Operators

7a. Use a set operator to combine multiple queries into a single query
7b. Control the order of rows returned

8. Using DDL Statements to Create and Manage Tables

8a. Describe data types that are available for columns
8b. Create a simple table
8c. Create constraints for tables
8d. Describe how schema objects work
8e. Execute a basic SELECT statement

9. Managing Objects with Data Dictionary Views

9a .Use the data dictionary views to research data on objects
9b. Query various data dictionary views

10. Managing Schema Objects

10a. Manage constraints
10b. Create and maintain indexes including invisible indexes and multiple indexes on the same columns
10c. Drop columns and set column UNUSED
10d. Perform flashback operations
10e. Create and use external tables

11. Using Data Definition Language (DDL)

11a. Describe the purpose of DDL
11b. Use DDL to manage tables and their relationships
11c. Explain the theoretical and physical aspects of a relational database

12. Defining SELECT Statements

12a. Identify the connection between an ERD and a database using SQL SELECT statements

13. Restricting and Sorting Data

13a. Use the ORDER BY clause to sort SQL query results
13b. Limit the rows that are retrieved by a query
13c. Sort the rows that are retrieved by a query
13d. Use ampersand substitution to restrict and sort output at runtime

14. Using Single-Row Functions to Customize Output

14a. Use various types of functions available in SQL
14b. Use conversion functions
14c. Use character, number, and date and analytical (PERCENTILE_CONT, STDDEV, LAG, LEAD) functions in SELECT statements

15. Reporting Aggregated Data Using the Group Functions

15a. Describe the use of group functions
15b. Group data by using the GROUP BY clause
15c. Include or exclude grouped rows by using the HAVING clause

16. Using Subqueries to Solve Queries

16a. Define subqueries
16b. Describe the types of problems subqueries can solve
16c. Describe the types of subqueries
16d. Use correlated subqueries
16e. Update and delete rows using correlated subqueries
16f. Use the EXISTS and NOT EXISTS operators
16g. Use the WITH clause
16h. Use single-row and multiple-row subqueries

17. Manipulating Data

17a. Insert rows into a table
17b. Update rows in a table
17c. Delete rows from a table
17d. Control transactions

18. Creating Other Schema Objects

18a. Create simple and complex views with visible/invisible columns
18b. Create, maintain and use sequences

19. Controlling User Access

19a. Differentiate system privileges from object privileges
19b. Grant privileges on tables and on a user
19c. Distinguish between privileges and roles

20. Manipulating Large Data Sets

20a. Describe the features of multitable INSERTs
20b. Merge rows in a table